

Integral form of continuity $\rightarrow$ review of G.E & thermo laws

$$\frac{\partial}{\partial t} \iiint \rho dV + \iint \rho \vec{V} \cdot \vec{n} dS = 0$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = 0 \text{ steady}$$

Sec 3.2, General 1-D, steady flow of a perfect gas

$\rightarrow$ Gas mixtures

$$\nabla \cdot (\rho \vec{V}) = 0 \text{ for 1-D } \frac{\partial(\rho u)}{\partial x} = 0$$

Mass: $\dot{m}_2 = \dot{m}_1 \stackrel{\text{def}}{=} \dot{m}$ {integrated form of Eqn. (3.1)}

Momentum: $\dot{m}_2 u_2 = \dot{m}_1 u_1 + p_2 A_2 + p_1 A_1 + F$ {integrated form of Eqn. (3.2),

where F represents the sum of the forces exerted on the flow by the surfaces of the device it interacts with, and any body forces (gravity, electrical, magnetic forces.)

From this equation, I_2

$$F = F_v + F_b$$

$$F = (\dot{m}_2 u_2 + p_2 A_2) - (\dot{m}_1 u_1 + p_1 A_1)$$

The group $(\dot{m}u + pA)$ is referred to as the impulse function: $I \stackrel{\text{def}}{=} (\dot{m}u + pA)$.

Therefore, $F = (I_2 - I_1)$ The force exerted on the device by the flow is of the same magnitude, but in the opposite direction:

$-F_{\text{on the CV}} = F_{\text{on device}} = (I_1 - I_2)$

Law of the energy equation:

$$\dot{m} \left(h_2 + \frac{u_2^2}{2} \right) = \dot{m} \left(h_1 + \frac{u_1^2}{2} \right) + \dot{Q} - \dot{p}_s \quad \text{{integrated form of Eqn. (3.3a), where}}$$

$\dot{Q}$ is the rate of heat addition into the control volume and $\dot{p}_s$ is the work done by the control volume per unit time (shaft power).

Dividing by $\dot{m}$,

$$h_{t2} = h_{t1} \quad \left(h_2 + \frac{u_2^2}{2} \right) = \left(h_1 + \frac{u_1^2}{2} \right) + \frac{\dot{Q}}{\dot{m}} - \frac{\dot{p}_s}{\dot{m}}$$

$$h_{t2} = h_{t1} \text{ if } \begin{cases} \dot{q} = 0 \\ \dot{w} = 0 \end{cases}$$

Introduce the heat added per unit mass (q) and work done per unit mass (w):

$$q \stackrel{\text{def}}{=} \frac{\dot{Q}}{\dot{m}} \text{ (J/kg)}$$

$$w \stackrel{\text{def}}{=} \frac{\dot{p}_s}{\dot{m}} \text{ (J/kg)}$$

$$h_{t2} = h_{t1} + q - w$$

I law can be written as $\left(h_2 + \frac{u_2^2}{2} \right) = \left(h_1 + \frac{u_1^2}{2} \right) + q - w$ {integrated form of Eqn. (3.3b).

II Law:

$$\dot{m}(s_2 - s_1) = \iint \dot{q} dA + S_{\text{gen}} \quad \text{entropy generation}$$

$\dot{m}(s_2 - s_1) \geq \iint \dot{q} dA$, {integrated form of Eqn. (2.11), in which $\dot{q}$ is the heat transfer per unit area per unit time}. Inequality applied for irreversible processes and equality applied for a reversible process. For an adiabatic process, $\dot{Q} = \dot{q} = 0$, and $s_2 \geq s_1$. For a reversible, adiabatic process (isentropic process), $s_2 = s_1$.

$$\begin{cases} \text{if } \dot{q} = 0 \xrightarrow{\text{adiabatic}} s_2 \geq s_1 \\ \text{if } \dot{q} = 0, S_{\text{gen}} = 0 \rightarrow s_2 = s_1 \end{cases}$$

Stagnation state

Stagnation state is defined as the state attained by a fluid on being brought to rest ($u = 0$), adiabatically, reversibly and without work. In conjunction with the I and II laws, this definition yields

$$\boxed{h_0 = h + \frac{u^2}{2}}$$

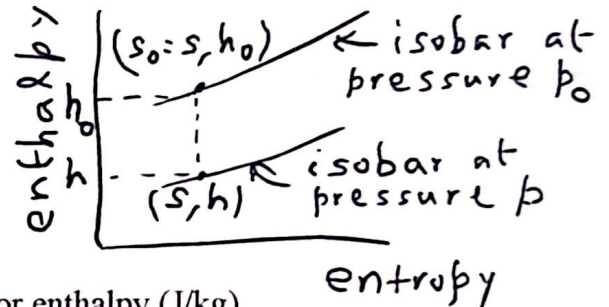
$$\boxed{s_0 = s}$$

An ideal gas with constant c_p or γ is said to be calorically perfect. For such a gas,

$$\frac{dh}{dT} = c_p \rightarrow h_0 - h = c_p(T_0 - T)$$

$$h_0 = h + \frac{u^2}{2} \text{ becomes } T_0 = T + \frac{u^2}{2c_p}$$

$$\text{Since } s_0 = s, \left(\frac{p_0}{p}\right) = \left(\frac{T_0}{T}\right)^{\frac{\gamma}{\gamma-1}}$$



h_0 – stagnation enthalpy (J/kg), h – static enthalpy, or enthalpy (J/kg)

s_0 – stagnation entropy (J/kg-K), s – static entropy, or entropy (J/kg-K)

T_0 – stagnation temperature (K), T – static temperature or temperature (K)

p_0 – stagnation pressure (Pa), p – static pressure, or pressure (Pa)

Summary

I Law:

$$\underbrace{\left(h_2 + \frac{u_2^2}{2}\right)}_{h_{02}} = \underbrace{\left(h_1 + \frac{u_1^2}{2}\right)}_{h_{01}} + q - w$$

$$\Delta h_0 = \rightarrow h_{02} - h_{01} = q - w$$

$$\rightarrow c_p(T_{02} - T_{01}) = q - w \quad \{\text{for a CPG}\}$$

II Law:

$$(s_2 - s_1) = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1}$$

$$\underbrace{s_{02} = s_2}, \underbrace{s_{01} = s_1} \text{ (definition of stagnation state)}$$

$$S_0 - S = c_p \ln \frac{T_0}{T} - R \ln \frac{P_0}{P}$$

$$S_0 = S \rightarrow \Delta S = 0 \Rightarrow \ln \left(\frac{T_0}{T}\right)^{c_p} = \ln \left(\frac{P_0}{P}\right)^R$$

$$\left(\frac{T_0}{T}\right)^{\frac{c_p}{R}} = \left(\frac{P_0}{P}\right)$$

$$\rightarrow (s_2 - s_1) = (s_{02} - s_{01}) = c_p \ln \frac{T_{02}}{T_{01}} - R \ln \frac{p_{02}}{p_{01}}$$

For isentropic flow from station (1) to station (2),

$$(s_2 - s_1) = 0 \rightarrow \left(\frac{p_2}{p_1} \right) = \left(\frac{T_2}{T_1} \right)^{\frac{\gamma}{\gamma-1}} \quad \{\text{relation between pressure ratio and temperature ratio in isentropic flow}\}$$

$$(s_{02} - s_{01}) = 0 \rightarrow \left(\frac{p_{02}}{p_{01}} \right) = \left(\frac{T_{02}}{T_{01}} \right)^{\frac{\gamma}{\gamma-1}} \quad \{\text{relation between stagnation pressure ratio and stagnation temperature ratio in isentropic flow}\}$$

Mach number

$M \stackrel{\text{def}}{=} \frac{u}{a}$, a is the local speed of sound $\rightarrow$ speed of propagation of information in gas

For an ideal gas, $a = \sqrt{\gamma RT}$ {see Eqn. (3.8)}

$M < 1$ (subsonic), $M > 1$ (supersonic), $M = 1$ (sonic)

$$T_0 = T + \frac{u^2}{2c_p} = T + \frac{M^2 a^2}{2c_p} = T + \frac{M^2 \gamma RT}{2 \frac{\gamma R}{\gamma-1}} = T \left(1 + \frac{\gamma-1}{2} M^2 \right)$$

$$\rightarrow \frac{T_0}{T} = \left(1 + \frac{\gamma-1}{2} M^2 \right)$$

{see Eqn. 3.10}

does not need isentropic assumption, just $q = w$ for isentropic flow = 0

$$\frac{p_0}{p} = \left(\frac{T_0}{T} \right)^{\frac{\gamma}{\gamma-1}} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma}{\gamma-1}}$$

{see Eqn. 3.11}

Mass flow rate in terms of stagnation values and Mach number:

$$\dot{m} = \rho u A = \frac{p}{RT} M \sqrt{\gamma RT} A = \left(\sqrt{\frac{\gamma}{R}} \frac{p}{\sqrt{T}} M A \right)$$

Substituting $T = \frac{T_0}{\left(1 + \frac{\gamma-1}{2} M^2 \right)}$ $p = \frac{p_0}{\left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma}{\gamma-1}}}$

$$\dot{m} = \sqrt{\frac{\gamma}{R}} \frac{p_0}{\sqrt{T_0}} M \left(1 + \frac{\gamma-1}{2} M^2 \right)^{-\frac{\gamma+1}{2(\gamma-1)}} A \quad \{\text{see Eqn. 3.13}\}$$

$$\dot{m} = \dot{m}(p_0, T_0, \gamma, R, A)$$

Examples:

- (1) The heat transfer out of a flow per unit mass is 200 kJ/kg. If the mass flow rate is 50 kg/s, what is the heat transfer rate?

$$\dot{Q} = \dot{m}q = (50)(-200) = -10000 \frac{\text{kJ}}{\text{s}} = -10000 \text{ kW}.$$

- (2) The shaft work done per unit mass on a flow is 150 kJ/kg. If the mass flow rate is 50 kg/s, what is the shaft power? Is the power delivered or absorbed?

$$\dot{p}_s = \dot{m}w = (50)(-150) = -7500 \frac{\text{kJ}}{\text{s}} = -7500 \text{ kW}.$$

The power is absorbed by the device.

- (3) Heat and work interactions occur as a flow moves from station (1) to station (2). $q = 200 \text{ kJ/kg}$. $w = 200 \text{ kJ/kg}$. What is the change in stagnation enthalpy of the fluid?

$$h_{02} - h_{01} = q - w = 200 - 200 = 0.$$

- (4) At a certain station in a flow, $h = 0.5 \text{ MJ/kg}$ and $u = 400 \frac{\text{m}}{\text{s}}$. What is the stagnation enthalpy there? Also, what is $(s_0 - s)$ there?

$$h_0 = h + \frac{u^2}{2} = (0.5)(10^6) + \frac{400^2}{2} = 5.8(10^5) \frac{\text{J}}{\text{kg}} = 0.58 \text{ MJ/kg}.$$

$$(s_0 - s) = 0 \text{ (by definition).}$$

- (5) At a certain station in the flow of a calorically perfect gas (CPG), $T = 300 \text{ K}$, $p = 120 \text{ kPa}$, and $u = 300 \frac{\text{m}}{\text{s}}$. Find the stagnation temperature and stagnation pressure at this station $\{c_p = 1107 \frac{\text{J}}{\text{kg-K}}, R = 287 \frac{\text{J}}{\text{kg-K}}\}$.

$$T_0 = T + \frac{u^2}{2c_p} = 300 + \frac{300^2}{(2)(1107)} = 340.7 \text{ K}$$

$$\gamma = \frac{c_p}{c_p - R} = \frac{1107}{1107 - 287} = 1.35$$

$$\frac{p_0}{p} = \left(\frac{T_0}{T}\right)^{\frac{\gamma}{\gamma-1}} = \left(\frac{340.7}{300}\right)^{\frac{1.35}{1.35-1}} = 1.6335$$

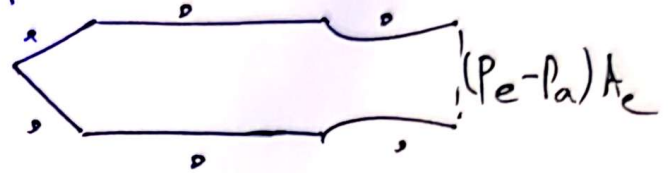
$$p_0 = 1.6335p = 1.6335(120) = 196 \text{ kPa}.$$

- (6) Hydrogen (H_2) is heated in a nuclear rocket heat exchanger from stagnation temperature 300 K to stagnation temperature 2300 K. Find q .

$$R_{\text{H}_2} = \frac{\bar{R}}{\bar{M}_{\text{H}_2}} = \frac{8314.3}{2} = 4157.2 \frac{\text{J}}{\text{kg-K}}$$

$$c_{p,\text{H}_2} = \frac{\gamma R_{\text{H}_2}}{\gamma - 1} = 18492 \frac{\text{J}}{\text{kg-K}}$$

As $\oint p_a dA = 0$, we reduce $\oint p_a dA$ from all the surfaces



$$c_p(T_{02} - T_{01}) = q - w = q - 0 = q$$

$$q = 18492(2300 - 300) = 3.6984(10^7) \frac{J}{kg} = 36984 \text{ kJ/kg}$$

(7) In a flow across a device, the stagnation temperature remains constant, and no work is done. The stagnation pressure drops from 7.8 atm to 5.6 atm. Find (a) the heat transfer per unit mass (q) and (b) the (specific) entropy change. ($\gamma = 1.4$, $R = 287 \text{ J/kg-K}$).

$$(a) c_p(T_{02} - T_{01}) = q - w = q - 0 = q$$

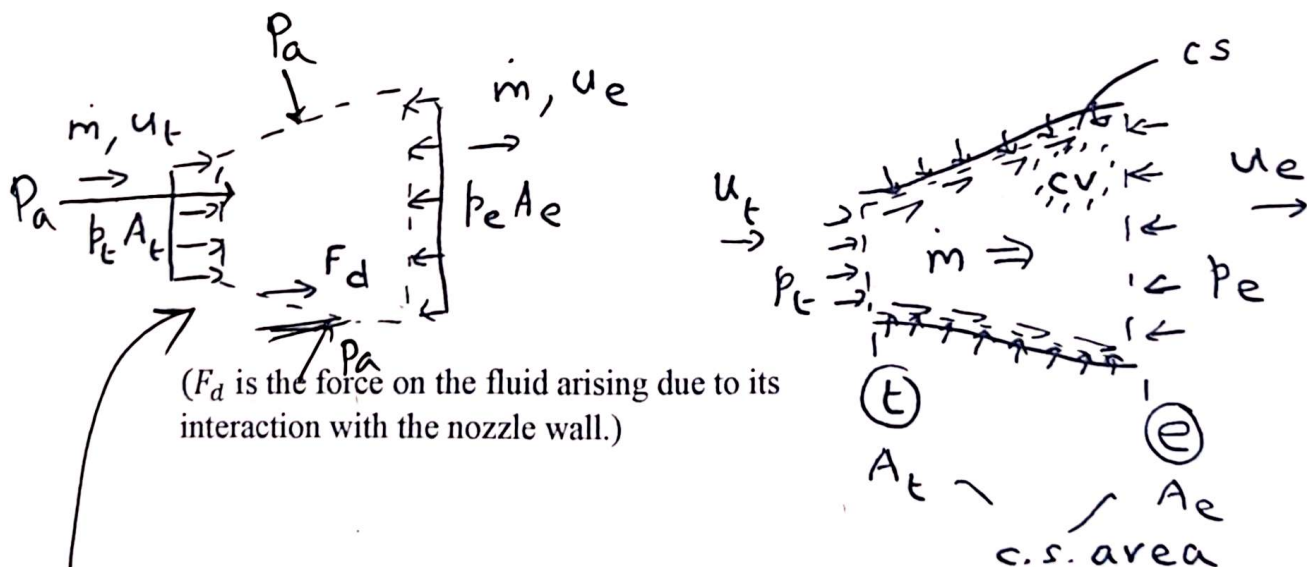
Since $T_{02} = T_{01}$, $q = 0$.

(b) $s_2 = s_{02}$, $s_1 = s_{01}$ (by definition of stagnation state)

$$\Delta s = (s_2 - s_1) = (s_{02} - s_{01}) = c_p \ln \frac{T_{02}}{T_{01}} - R \ln \frac{p_{02}}{p_{01}} = c_p \ln 1 - 287 \ln \frac{5.6}{7.8}$$

$$= 0 - (-95.1) = 95.1 \frac{J}{kg-K}$$

(8) Consider the diverging section of a rocket nozzle, see Figure. Derive the equation for the thrust developed by the nozzle. Show the direction of action of this force.



(F_d is the force on the fluid arising due to its interaction with the nozzle wall.)

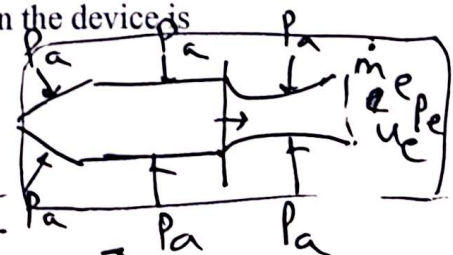
Momentum equation from (t) to (e):

$$\dot{m}u_t + p_t A_t + F_d - p_e A_e = \dot{m}u_e$$

Or $F_d = (\dot{m}u_e + p_e A_e) - (\dot{m}u_t + p_t A_t) = (I_e - I_t)$ This is the force exerted by the nozzle wall on the fluid. The force exerted by the fluid on the device is the same, but in the opposite direction ($\leftarrow$).

$$T = \dot{m}u_e + (p_e - p_a)A_e$$

we know $\oint p_a dA = 0$
that $\Rightarrow$



Ideal Gas Mixtures

Each constituent of a mixture is referred to as a species.

An ideal gas mixture, at a temperature T_m , in a container of volume V_m , has the following characteristics:

$$T_m = T_1 = T_2 = \dots = T_n \quad (2.23 a) \quad T = \frac{1}{N} \sum_{i=1}^N \frac{1}{2} m \bar{v}_i^2$$

$$p_m = p_1 + p_2 + \dots + p_n = \sum_{i=1}^n p_i \quad (2.23 b) \quad N = \text{number of molecules}$$

$$V_m = V_1 = V_2 = \dots = V_n \quad (2.23 c)$$

$$E_m = E_1 + E_2 + \dots + E_n = \sum_{i=1}^n E_i \quad (2.23 d)$$

$$H_m = H_1 + H_2 + \dots + H_n = \sum_{i=1}^n H_i \quad (2.23 f) \rightarrow H = C_p T$$

$$S_m = S_1 + S_2 + \dots + S_n = \sum_{i=1}^n S_i \quad (2.23 e) \quad H = \frac{m}{\lambda} C_p \frac{T}{\lambda}$$

The quantities $p_1, p_2, \dots, p_n$ in (2.23 b) are known as partial pressures. The partial pressure would be the pressure exerted by a species were it to occupy the volume V_m all by itself at the temperature T_m ; that is,

$$p_i(V_m) = n_i R T_m$$

Writing the above equation for the various species,

$$p_1 v_m = n_1 \bar{R} T_m \quad \begin{cases} p_1 - \text{partial pressure of species 1.} \\ n_1 - \text{number of moles of species 1 (kmol)} \end{cases}$$

$$p_2 v_m = n_2 \bar{R} T_m$$

⋮

$$p_n v_m = n_n \bar{R} T_m$$

Add: $\underbrace{(p_1 + p_2 + \dots + p_n)}_{p_m, \text{ see (2-23 b)}} v_m = \underbrace{(n_1 + n_2 + \dots + n_n)}_{n_m \text{ \# of moles in mixture}} \bar{R} T_m$

$$\boxed{p_m v_m = n_m \bar{R} T_m} \leftarrow \text{A mixture of ideal gases also behaves as an ideal gas}$$

$$\Rightarrow \frac{p_m v_m}{m_m} = \frac{n_m}{m_m} \bar{R} T_m \quad \begin{matrix} \leftarrow \text{mixture mass} \\ m_m = (m_1 + m_2 + \dots + m_n) \end{matrix}$$

$$\Rightarrow \frac{p_m}{\rho_m} = \frac{1}{\bar{M}_m} \bar{R} T_m = R_m T_m, \quad \begin{matrix} R_m = \frac{\bar{R}}{\bar{M}_m} \\ \uparrow \\ \text{mixture gas constant} \end{matrix}$$

$$\boxed{p_m = \rho_m R_m T_m}$$

the mixture

$$p = \rho R T$$

Thermal equation of state for species i : $p_i v_m = n_i \bar{R} T_m$

Thermal equation of state for mixture: $p_m v_m = n_m \bar{R} T_m$

$$\Rightarrow \frac{p_i}{p_m} = \frac{n_i}{n_m} = X_i$$

$$p_m = \rho_m R_m T_m$$

$\frac{n_i}{n_m}$ is called the mole fraction, and is designated X_i

$$\underline{X_i} \stackrel{\text{def}}{=} \frac{n_i}{n_m} \quad (\text{mole fraction of species } i)$$

$$\Rightarrow \frac{p_i}{p_m} = X_i \Rightarrow p_i = \underline{X_i p_m} \quad \text{see Eqn (2.25)}$$

$$m_m = m_1 + m_2 + \dots + m_n = \sum_{i=1}^n m_i$$

molar mass $\leftarrow$ number of moles

$$= \bar{M}_1 n_1 + \bar{M}_2 n_2 + \dots + \bar{M}_n n_n = \sum_{i=1}^n (\bar{M}_i n_i)$$

$$\bar{M}_m = \frac{m_m}{n_m} = \frac{\bar{M}_1 n_1 + \bar{M}_2 n_2 + \dots + \bar{M}_n n_n}{n_m}$$

$$= \bar{M}_1 X_1 + \bar{M}_2 X_2 + \dots + \bar{M}_n X_n = \sum_{i=1}^n (X_i \bar{M}_i)$$

$$\boxed{\bar{M}_m = \sum_{i=1}^n (X_i \bar{M}_i)}$$

$$(2.23f): H_m = H_1 + H_2 + \dots + H_n$$

$$= \underset{\substack{\text{number} \\ \text{of moles}}} {n_1} \underset{\substack{\text{molar} \\ \text{enthalpy}}} {\bar{h}_1} + n_2 \bar{h}_2 + \dots + n_n \bar{h}_n = \frac{\sum_{i=1}^n (n_i \bar{h}_i)}{n_m}$$

$$\bar{h}_m = \frac{H_m}{n_m} = \frac{n_1 \bar{h}_1 + n_2 \bar{h}_2 + \dots + n_n \bar{h}_n}{n_m}$$

$$= x_1 \bar{h}_1 + x_2 \bar{h}_2 + \dots + x_n \bar{h}_n = \boxed{\sum_{i=1}^n (x_i \bar{h}_i)}$$

$$\bar{h}_m = \sum_{i=1}^n (x_i \bar{h}_i)$$

Similarly,

$$\left\{ \begin{array}{l} \bar{s}_m = \sum_{i=1}^n (x_i \bar{s}_i) \\ \bar{c}_{pm} = \sum_{i=1}^n (x_i \bar{c}_{pi}) \\ \gamma_m = \frac{\bar{c}_{pm}}{\bar{c}_{vm}} = \frac{\bar{c}_{pm}}{\bar{c}_{pm} - \bar{R}} \end{array} \right.$$

A gas mixture can be simply treated as an ideal gas with molar mass $\bar{M}_m$ and specific heat ratio γ_m .

From $\bar{M}_m$, the gas constant of the mixture is obtained

$$\text{as } \bar{R}_m = \frac{\bar{R}}{\bar{M}_m}$$

$$X_i = \frac{n_i}{n_m}$$

Ex

The ideal gas product mixture formed by the combustion of H_2 and O_2 has the following characteristics:

species (M)	# mol (kmol)	$\bar{c}_p \left(\frac{kJ}{kmol \cdot K} \right)$	$\bar{h} \left(\frac{kJ}{kmol} \right)$	Calculated X_i
O_2 (32)	0.1008	36.3	116233	0.0433
H_2 (2)	0.3170	32.9	105346	0.1363
O (16)	0.054	20.9	312065	0.0232
H (1)	0.109	20.9	283015	0.0469
OH (17)	0.233	33.5	149137	0.1002
H_2O (18)	1.512	48.27	-87266	0.6501

Calculate the total number of moles in the mixture the species mole fractions, the mixture specific gas constant, and its molar-specific and mass-specific enthalpies. Also find the specific heat ratio of the mixture

$$\underline{n_m} = \sum n_i = \underline{2.3258}, \quad X_i = \frac{n_i}{n_m}, \text{ see table}$$

$$\bar{M}_m = \sum_{i=1}^n (X_i \bar{M}_i) = 15.48 \frac{\text{kg}}{\text{kmol}}$$

$$\bar{R} = 8.3143 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

$$R_m = \frac{\bar{R}}{\bar{M}_m} = \frac{8314.3}{15.48} = 537 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

$$\bar{h}_m = \sum_{i=1}^n (X_i \bar{h}_i) = -1883 \frac{\text{kJ}}{\text{kmol}}$$

$$\bar{h}_m = \frac{\bar{h}_m}{\bar{M}_m} = \frac{-1883}{15.48} = -121.7 \frac{\text{kJ}}{\text{kg}}$$

$$\bar{c}_{pm} = \sum_{i=1}^n X_i \bar{c}_{pi} = 42.258 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

$$\gamma_m = \frac{\bar{c}_{pm}}{\bar{c}_{pm} - \bar{R}} = \frac{42258}{42258 - 8314.3} = 1.24$$

$$\gamma_{\text{air}} = 1.4$$

↓ If $\bar{c}_{pi}$ values are not given, the following may be assumed for a preliminary calculation.

$$\gamma_{\text{mon}} = 1.67, \gamma_{\text{di}} = 1.29, \gamma_{\text{tri}} = 1.17$$

In this case, get $\bar{c}_{pi}$ from $\bar{c}_{pi} = \frac{\gamma_i \bar{R}}{\gamma_i - 1}$. Then

$$\text{get } \bar{c}_{pm} = \sum_{i=1}^n X_i \bar{c}_{pi}, \text{ and } \gamma_m = \frac{\bar{c}_{pm}}{\bar{c}_{pm} - \bar{R}}$$

$$c_p = \frac{\gamma R}{\gamma - 1} \Rightarrow c_{pi} = \frac{\gamma_i \bar{R}}{\gamma_i - 1}$$

Ex: Reconsider previous problem. Calculate γ_m using the γ -approximations for the species.

$$\bar{c}_{pi} = \frac{\gamma_i \bar{R}}{\gamma_i - 1}$$

$$\bar{c}_{p_{O_2}} = \bar{c}_{p_{H_2}} = \bar{c}_{p_{OH}} = \frac{\gamma_{di} \bar{R}}{\gamma_{di} - 1} = \frac{1.29 \bar{R}}{1.29 - 1} = 4.448 \bar{R}$$

$$\bar{c}_{p_O} = \bar{c}_{p_H} = \frac{\gamma_{mon} \bar{R}}{\gamma_{mon} - 1} = \frac{1.67 \bar{R}}{1.67 - 1} = 2.493 \bar{R}$$

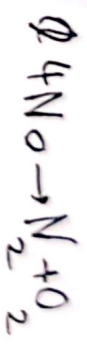
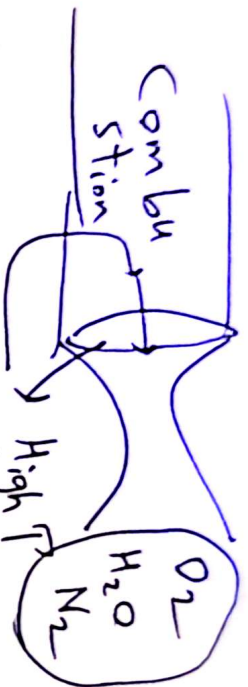
$$\bar{c}_{p_{H_2O}} = \frac{\gamma_{tri} \bar{R}}{\gamma_{tri} - 1} = \frac{1.17 \bar{R}}{1.17 - 1} = 6.882 \bar{R}$$

$$\bar{c}_{pm} = \sum_{i=1}^n (X_i \bar{c}_{pi}) = 5.893 \bar{R} = 5.893 \times 8.3143 \frac{kJ}{kmol \cdot K} = 48.996 \frac{kJ}{kmol \cdot K}$$

$$\gamma_m = \frac{\bar{c}_{pm}}{\bar{c}_{pm} - \bar{R}} = \frac{5.893 \bar{R}}{5.893 \bar{R} - \bar{R}} = \frac{5.893 \bar{R}}{4.893 \bar{R}} = 1.20$$

1.2 vs 1.17

not bad!



Ex: A mixture of gases enters a rocket nozzle with the composition shown in the table, below:

Species	H ₂	O ₂	H ₂ O	OH	O	H	N ₂	N	NO	NO ₂
Mole fraction	0.0215	0.1518	0.4473	0.0786	0.01832	0.00847	0.2382	8.35(10 ⁻⁴)	0.03578	9.77(10 ⁻⁴)

As it flows through the nozzle, its composition changes due to recombination processes.

The mixture at the nozzle exit consists only of the species O₂, H₂O and N₂. Determine the specific gas constant and the specific heat ratio of the mixture at the nozzle exit. $\gamma_{mon} = 1.67$, $\gamma_{di} = 1.29$, $\gamma_{tri} = 1.17$.

Assume that the number of moles of the mixture at the nozzle entrance to be n_m . Therefore, the number of moles of the various species are

$$\begin{aligned} n_{\text{H}_2} &= 0.0215n_m, \quad n_{\text{O}_2} = 0.1518n_m, \quad n_{\text{H}_2\text{O}} = 0.4473n_m, \quad n_{\text{OH}} = 0.0786n_m, \quad n_{\text{O}} = 0.01832n_m, \\ n_{\text{H}} &= 0.00847n_m, \quad n_{\text{N}_2} = 0.2382n_m, \quad n_{\text{N}} = 8.35(10^{-4})n_m, \quad n_{\text{NO}} = 0.03578n_m, \\ n_{\text{NO}_2} &= 9.77(10^{-4})n_m. \end{aligned}$$

Now consider the mixture at the nozzle exit. Let the number of moles of the species O₂, H₂O and N₂ be a , b and c , respectively.

Atoms are conserved in chemical reactions. Therefore, the number of atoms of O, H and N should be the same at the nozzle entrance and the nozzle exit.

Equating the number of atoms of N,

$$2c = 2n_{\text{N}_2} + n_{\text{N}} + n_{\text{NO}} + n_{\text{NO}_2} = 0.51229n_m \rightarrow c = 0.2561n_m$$

Equating the number of atoms of H,

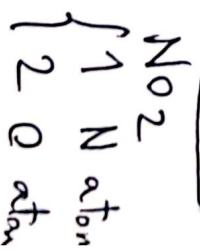
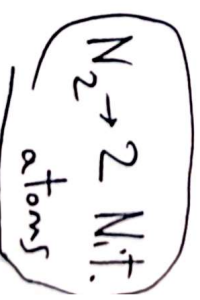
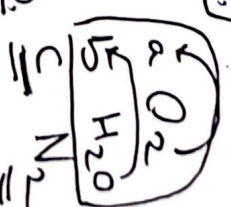
$$2b = 2n_{\text{H}_2} + 2n_{\text{H}_2\text{O}} + n_{\text{OH}} + n_{\text{H}} = 1.02467n_m \rightarrow b = 0.5123n_m$$

Equating the number of atoms of O,

$$2a + b = 2n_{\text{O}_2} + n_{\text{H}_2\text{O}} + n_{\text{OH}} + n_{\text{O}} + n_{\text{NO}} + 2n_{\text{NO}_2} = 0.8838n_m \rightarrow a = 0.1858n_m$$

Remembering that a , b and c are the number of moles, the total number of moles of mixture at station 2 is $n_{m2} = a + b + c = 0.9542n_m$.

Observe: The total number of moles of species at the entrance (n_m) is not the same as that at the exit ($0.9542n_m$); that is, the number of moles is not conserved in a reaction. There are some chemical reactions in which the number of moles is conserved, but this does not generally happen.



n_m

$0.9542 n_m$

In any event, $X_{O2} = a/n_{m2} = 0.1947$, $X_{H2O} = b/n_{m2} = 0.5369$, $X_{N2} = c/n_{m2} = 0.2684$.

$$\bar{M}_{m2} = \sum(X_i \bar{M}_i) = 23.41$$

$$R_{m2} = \frac{\bar{R}}{\bar{M}_{m2}} = 355.16 \text{ J/kg-K}$$

$$\bar{c}_{pm2} = \sum(X_i \bar{c}_{pi}) = X_{O2} \bar{c}_{pO2} + X_{H2O} \bar{c}_{pH2O} + X_{N2} \bar{c}_{pN2}$$

$$\bar{c}_{pO2} = \bar{c}_{pN2} = \frac{\gamma_{di} \bar{R}}{\gamma_{di} - 1} = 4.4483 \bar{R}$$

$$\bar{c}_{pH2O} = \frac{\gamma_{tri} \bar{R}}{\gamma_{tri} - 1} = 6.8824 \bar{R}$$

$$\bar{c}_{pm2} = (0.1947)(4.4483 \bar{R}) + (0.2684)(4.4483 \bar{R}) + (0.5369)(6.8824 \bar{R}) = 5.7552 \bar{R}$$

$$\gamma_{m2} = \frac{\bar{c}_{pm2}}{\bar{c}_{pm2} - \bar{R}} = 1.21$$